

Enterprise VLAN & Trunking Lab

Phase 6 Report – Network Troubleshooting & Validation

Project Information

Project Name: Enterprise VLAN & Trunking Lab

Phase: Phase 6 – Network Troubleshooting & Validation

Simulation Software: Cisco Packet Tracer

Devices Verified: SW1, SW2, Eight End Devices

Status: Completed

1. Introduction

After validating the correct operation of the enterprise Layer 2 network in Phase 5, the final technical phase focused on troubleshooting and validation. Rather than deploying new features, this phase simulated common network failures encountered in production environments and applied a structured troubleshooting methodology to identify and resolve them.

Troubleshooting is a fundamental responsibility of network engineers. The ability to diagnose network problems systematically minimizes downtime, ensures service availability, and improves operational reliability. Throughout this phase, several Layer 2 issues were intentionally introduced, analyzed, corrected, and verified using Cisco IOS diagnostic commands.

2. Objectives

The objectives of this phase were to:

- Apply a structured network troubleshooting methodology.
- Diagnose common VLAN configuration errors.
- Identify trunk configuration problems.
- Detect native VLAN mismatches.
- Verify interface operational status.
- Analyze MAC address tables.
- Restore normal network operation after configuration errors.
- Validate the corrected network configuration.

3. Troubleshooting Methodology

A standardized troubleshooting process was followed for every scenario.

```
graph TD; A[Problem Identification] --> B[Information Collection]; B --> C[Root Cause Analysis]; C --> D[Corrective Action]; D --> E[Verification]; E --> F[Documentation];
```

This systematic approach ensures efficient fault isolation while minimizing unnecessary configuration changes.

4. Troubleshooting Scenario 1 – Incorrect VLAN Assignment

Problem Description

A Management department workstation was unable to communicate with another workstation within the same VLAN.

Investigation revealed that the switch interface connected to the workstation had been assigned to VLAN 20 instead of VLAN 10.

Diagnostic Procedure

The following command was executed:

```
show vlan brief
```

The output indicated that the affected interface was associated with the wrong VLAN.

Root Cause

Incorrect access VLAN assignment.

The interface was configured as:

- VLAN 20

instead of:

- VLAN 10
-

Corrective Action

The interface was reassigned to VLAN 10 using Cisco IOS configuration commands.

Verification

The VLAN database was reviewed again using:

```
show vlan brief
```

Connectivity testing confirmed successful communication between Management hosts.

5. Troubleshooting Scenario 2 – Trunk Misconfiguration

Problem Description

The inter-switch connection failed to operate as a trunk because the interface had been configured as an access port.

Diagnostic Procedure

The following command was executed:

```
show interfaces trunk
```

The output indicated that no interfaces were operating in trunk mode.

Further verification was performed using:

```
show interfaces gigabitEthernet0/1 switchport
```

The administrative mode was identified as:

- Static Access
-

Root Cause

The interface had been incorrectly configured as an access port.

Corrective Action

The interface was reconfigured as an IEEE 802.1Q trunk.

The following parameters were restored:

- Trunk mode
 - Native VLAN 99
 - Allowed VLAN list
-

Verification

The following command confirmed successful trunk establishment:

```
show interfaces trunk
```

The trunk returned to an operational state.

6. Troubleshooting Scenario 3 – Native VLAN Mismatch

Problem Description

Different native VLANs were configured on the two switches.

SW1

- Native VLAN 99

SW2

- Native VLAN 1
-

Diagnostic Procedure

The trunk configuration was examined on both switches using:

```
show interfaces trunk
```

Comparison of the outputs revealed inconsistent native VLAN assignments.

Root Cause

Native VLAN mismatch between interconnected trunk ports.

Corrective Action

Both trunk interfaces were configured to use VLAN 99 as the native VLAN.

Verification

Subsequent verification confirmed that both switches shared the same native VLAN configuration.

7. Troubleshooting Scenario 4 – Missing Allowed VLAN

Problem Description

One VLAN was accidentally removed from the trunk's allowed VLAN list.

As a result, traffic belonging to that VLAN would be unable to traverse the trunk if the VLAN existed on both switches.

Diagnostic Procedure

The allowed VLAN list was verified using:

```
show interfaces trunk
```

The output revealed that VLAN 20 was absent from the list.

Root Cause

Incorrect trunk allowed VLAN configuration.

Corrective Action

The missing VLAN was restored to the allowed VLAN list.

Verification

The updated trunk configuration confirmed that all required VLANs were permitted across the trunk.

8. Troubleshooting Scenario 5 – Interface Shutdown

Problem Description

The inter-switch trunk interface had been administratively disabled.

Diagnostic Procedure

Interface status was examined using:

```
show interfaces status
```

The interface state appeared as:

- Disabled
-

Root Cause

The interface had been manually shut down.

Corrective Action

The interface was re-enabled using the appropriate Cisco IOS command.

Verification

Interface status returned to:

- Connected

Normal network operation was restored.

9. Troubleshooting Scenario 6 – Empty MAC Address Table

Problem Description

The MAC address table initially contained no dynamically learned addresses.

Diagnostic Procedure

The following command was executed:

```
show mac address-table
```

No dynamic entries were displayed.

Root Cause

No Layer 2 traffic had been generated since the switch was powered on.

The switch therefore had not yet learned any source MAC addresses.

Corrective Action

Traffic was generated by initiating communication between hosts within the same VLAN.

Verification

The MAC address table was examined again.

Dynamic MAC address entries appeared, confirming successful address learning by the switch.

10. Validation Results

After resolving each troubleshooting scenario, the complete network was revalidated.

The following components were confirmed to operate correctly:

Component	Status
VLAN Configuration	Verified
Access Port Assignment	Verified
Trunk Configuration	Verified
Native VLAN	Verified
Allowed VLAN List	Verified
Interface Status	Verified
MAC Address Learning	Verified
Intra-VLAN Connectivity	Verified
VLAN Isolation	Verified
Running Configuration	Verified

11. Cisco IOS Diagnostic Commands Used

The following commands were used throughout the troubleshooting process:

Command	Purpose
<code>show vlan brief</code>	Verify VLAN membership
<code>show interfaces trunk</code>	Verify trunk operation
<code>show interfaces status</code>	Check interface status
<code>show interfaces gigabitEthernet0/1 switchport</code>	Verify switchport mode
<code>show mac address-table</code>	Display learned MAC addresses
<code>show running-config</code>	Review switch configuration
<code>ping</code>	Test connectivity
<code>copy running-config startup-config</code>	Save configuration

12. Technical Concepts Reinforced

This phase strengthened practical understanding of:

- Layer 2 troubleshooting methodology
- VLAN fault isolation
- IEEE 802.1Q trunk troubleshooting
- Native VLAN mismatch detection
- Interface diagnostics
- MAC address learning analysis
- Enterprise network validation
- Cisco IOS verification procedures

These skills represent core competencies expected of professional network engineers responsible for maintaining enterprise switching infrastructures.

13. Best Practices Applied

The following operational best practices were followed:

- Troubleshoot before modifying configurations.
- Verify every change after implementation.
- Use diagnostic commands to confirm root causes.
- Restore configurations according to the original network design.
- Save verified configurations.
- Document every issue and its resolution.

Following these practices reduces operational risk and improves network reliability.

14. Lessons Learned

Several important networking principles were demonstrated during this phase:

- Most network problems can be diagnosed using Cisco IOS verification commands.
 - Incorrect VLAN assignments directly affect Layer 2 connectivity.
 - Trunk misconfigurations can interrupt communication between switches.
 - Native VLAN consistency is essential for stable trunk operation.
 - MAC address tables provide valuable information for troubleshooting switching behavior.
 - A structured troubleshooting methodology is more effective than trial-and-error configuration changes.
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15. Skills Developed

Upon completion of this phase, the following practical skills were acquired:

- Diagnosing VLAN configuration issues.
- Troubleshooting trunk links.
- Identifying interface configuration errors.
- Detecting native VLAN mismatches.
- Analyzing MAC address tables.
- Validating enterprise switch configurations.
- Applying structured fault isolation techniques.
- Restoring network services efficiently.

These skills are directly applicable to enterprise campus network administration and technical support roles.

16. Preparation for Phase 7

The technical implementation of the enterprise switching project has now been completed.

The final phase will focus on professional project presentation, including:

- Organizing the project directory.
 - Preparing GitHub documentation.
 - Writing a comprehensive README.
 - Creating network diagrams.
 - Collecting configuration files.
 - Adding screenshots and verification evidence.
 - Finalizing the project as a portfolio-ready showcase.
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17. Conclusion

Phase 6 successfully demonstrated the practical application of enterprise network troubleshooting techniques. Multiple Layer 2 faults were intentionally introduced, diagnosed, corrected, and verified using Cisco IOS diagnostic commands and a structured troubleshooting methodology.

The successful resolution of these scenarios confirms not only the correctness of the network implementation but also the ability to identify and resolve operational issues commonly encountered in production environments. Completing this phase strengthens the practical troubleshooting skills required of network engineers and prepares the project for its final documentation and presentation stage.